

Math 105LA Assignment 4

Fall 2019

Due Sun 10/27, 11pm

Write a MATLAB function to implement Gaussian Elimination with partial pivoting.

Input: augmented matrix, A

Output: solution of the system, x

Step 1: Set n = number of rows of A

```
For i = 1,...,n,  
    set NROW(i) = i;
```

Step 2: For i = 1,2,...,n-1, repeat step 3- step 6

Step 3: Let p >= i be the smallest integer with
 $|A(NROW(p), i)| = \max_{\{i \leq j \leq n\}} \{|A(NROW(j), i)|\}$

Step 4: If A(NROW(p), i) = 0
Output('no unique solution exists'); stop

Step 5: If p not equal to i
NCOPY = NROW(i), NROW(i) = NROW(p), NROW(p) = NCOPY

Step 6: For j = i+1, i+2, ..., n
set m = A(NROW(j), i) / A(NROW(i), i),
set A(NROW(j), :) = A(NROW(j), :) - m * A(NROW(i), :)

Step 7: If A(NROW(n), n) = 0
Output('no unique solution exists'); stop

Step 8: Set x(n) = A(NROW(n), n+1) / A(NROW(n), n)

Step 9: For i = n-1, n-2, ..., 1
set x(i) = (A(NROW(i), n+1) - $\sum_{j=i+1}^n A(NROW(i), j)x(j)$) / A(NROW(i), i)

Step 10: Output(x), stop

You can start with the template below. Make sure that you use the **same order** of input:

```
function x = gaussian(A)  
% A = augmented matrix of a linear system  
% x = the solution of the linear system  
end
```

For example, to solve linear system

$$\begin{cases} x_1 - 2x_2 + 3x_3 = 1 \\ 2x_1 - 4x_2 + 5x_3 = 0 \\ -x_1 + x_2 + x_3 = -1 \end{cases}$$

First define the augmented matrix:

```
A = [1 -2 3 1; 2 -4 5 0; -1 1 1 -1];
```

Recall that the semicolon indicates the start of another row. Then, assuming your function is saved in 'gaussian.m', type the following in the command window (without the semicolon):

```
x = gaussian(A)
```

And your function should return

```
x =
    11
     8
     2
```

One can verify that $A(:, 1:\text{end}-1) * x$ should produce the numbers on the right hand side. Here $A(:, 1:\text{end}-1)$ cuts off the last column of A and is the coefficient matrix of the linear system.

More examples:

Define ...	Then call your algorithm ...	And it should return ...
A = [4 -1 1 8; 2 5 2 3; 1 2 4 11];	gaussian(A)	1 -1 3
A = [1 -1 2 -1 -3; 2 -2 3 -3 -20; 1 1 1 0 -2; 1 -1 4 3 4];	gaussian(A)	-44.5000 18.0000 24.5000 -10.5000
A = [1, 0, 1 1; 0, 1e-16, 3, 1; 1, 1, 1, 3]	gaussian(A)	0.6667 2.0000 0.3333
A = [1 1 0 1 2; 2 1 -1 1 1; 4 -1 -2 1 0; 3 -1 -1 2 3];	gaussian(A)	0 0 1 2

Submit a single file **named by your UCInetID** (e.g. wtleung.m) that contains the function. Your code will be tested on a few problems. Your grade of this assignment is calculated by the number of problems that your code solves correctly. Make sure your function runs without errors and has **the same order of input** above. **Failure to do so will result in a zero grade.**

You can use matlab function “size” to output the number of rows (or columns) of A

Example:

```
A = [1, 2, 4, 5; 7, 8];
```

size(A,1) = 3 is the number of rows

size(A,2) = 2 is the number of columns

In matlab, $A(i, j : j+k)$ represent a row vector $[a_{i,j+1}, a_{i,j+2}, \dots, a_{i,j+k}]$

Similarly, $A(j : j+k, i)$ represent a column vector $[a_{j+1,i}, a_{j+2,i}, \dots, a_{j+k,i}]$

Example:

`A = [1, 2, 3; 4, 5, 6; 7, 8, 9];`

`A(2,2:3)` is [5, 6]

`A(1:3,1)` is [1; 2; 3]

You can use matlab function “max” to output the maximum of a vector and the position corresponding to the maximum value.

Example:

`v = [1; 5; 3; 2]`

`[m,p] = max(v)`

The maximum value, m is 5

The position, p is 2